INCH-POUND

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MILITARY SPECIFICATION

ELECTRO-CHEMICAL MACHINE, GRINDING, SURFACE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electrolytic surface grinding machine, single end, floor mounted, motor driven, horizontal spindle, complete with equipment and accessories as specified herein.

1.2 Classification. The electrolytic surface grinding machine shall be of the following sizes. The sizes required shall be as specified (see 6.2).

Size 1	6" x 18" worktable
Size 2	8" x 24" worktable
Size 3	12" x 36" worktable

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3410

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 218.01 - Standard for Rating the Pitting Resistance and Bending Strength of Spur and Helical Involute Gear Teeth.

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

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JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG-1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4 and 6.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current commercial production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

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3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Digital readout system. When specified (see 6.2), a digital readout system shall be provided on the X and Y axes of travel. The digital readout system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0001 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0002 inch, measured over the full travel on each axis. Special requirements for the digital readout system components shall be as specified (see 6.2). The digital readout shall indicate axis location in both the US and SI units. The digital readout system shall be provided with electronic and manual calibration to the machine on which it is installed to eliminate the effects of distortion error.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines procured under the same contract.

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3.2.1.2.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's normal work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of installed cables shall prevent any interference with normal machine operation. Electronic display consoles shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for the point of operation and all other parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, (see 6.2) shall comply with all of the requirements of OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

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3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine and its components, materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. Means shall be provided to indicate a low oil condition. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. Replacement parts shall be domestically available.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

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3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 218.01, and AGMA 2000-A88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than the following components.

3.4.1 Base. The base shall be a casting of iron or iron alloy, or a fabricated steel weldment, designed for floor mounting. The structure shall be stress relieved and normalized to minimize distortion and deflection. If drive motors, electrolyte reservoir or other internal mechanisms are mounted in the base, doors or readily removable covers shall be provided for servicing the components. The base shall incorporate, either as an integral part or as a separate unit, provisions for accepting the saddle and sliding table. Means shall be provided on the base for leveling.

3.4.2 Saddle. The saddle shall be a casting of iron or iron alloy and shall be designed and proportioned to withstand all loads which may be imposed in normal operation of the machine. The saddle shall be mounted on the guideways provided on the base structure. Ways or guides shall be provided at the top of the saddle to incorporate a sliding table, and shall be covered with a polymer, plastic, or resin having a formula or (C2F4)N. The guideways and feed mechanism shall be constructed to such precision as to allow positioning the saddle to the accuracy specified in table II.

3.4.3. Table. The table shall be fabricated of corrosion-resistant steel or nickel plated iron or iron alloy. The table shall be mounted on ways or guides on the saddle and travel on the saddle in a direction square to the movement of the saddle movement. The table feed mechanism shall be constructed to such precision as to allow positioning of the table to the accuracy specified in table II. The number of T-slots shall be as specified in table I, and meet the requirements of ANSI B5.1. The accuracy of table surface, table travel, and table top incline shall conform to the accuracies specified in table II. Means shall be provided on the table to electrically insulate work holding fixtures or magnetic chucks, and cable leads shall be provided to connect fixtures or chucks to positive polarity.

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3.4.3.1 Table drive. The table drive shall be designed for providing rapid approach and controlled grinding feed rate as specified in table I. Limit switches and adjustable stops shall be mounted on the machine to provide table positioning and sequence control. A feed rate indicating meter shall be provided.

3.4.4 Column. The column shall be a casting of iron or iron alloy mounted on the base of the machine, and shall be designed for support of a grinding spindle.

3.4.5 Spindle. The spindle shall rotate clockwise on permanently lubricated bearings, shall be dynamically balanced, equipped with power pickup for electrolytic grinding and shall be electrically insulated from other parts of the grinder. The spindle shall be stainless steel, nickel plated, chromium plated or other simular corrosion resistant material. The spindle nose shall be tapered and shall accept wheel adapters with matching tapers. Provisions shall be made for ready removability and interchangeability of wheel adapters. The spindle shall be machined to such accuracy as to comply with wheel spindle taper and collet face runout as specified in table II.

3.4.6 Spindle drive. The spindle drive shall be either direct, as an integral part of the motor shaft, or by belt from the motor to the spindle. When belt drive is used, means shall be provided for adjustment of the belt tension. When direct drive is used, the electric current shall be supplied to the motor through a grounded isolation transformer. Motor horsepower shall correspond to those designated in table I.

3.4.7 Worktable enclosure. The worktable enclosure shall be provided to contain all spray and splash of electrolyte during grinding operation. The sides and top of the enclosure shall contain windows fabricated of an optically clear material which is not adversely affected by the electrolyte solution. Sliding access doors shall be fabricated of the same material as the windows. Means shall be provided to prevent dripping of electrolyte outside the enclosure when doors are opened. All metal portions of the enclosure shall be painted with an epoxy system and hardware shall be commercially available stainless steel, nickel plated, or otherwise suitably corrosion-protected.

3.4.8 Electrolyte system. The electrolyte system shall consist of a reservoir and pumping system to supply electrolyte to the grinding wheel and work piece, a filtering system, a complete mist collection system, and all necessary piping, hoses, valves, and meters.

3.4.8.1 Electrolyte reservoir. The electrolyte reservoir and its components shall be of AISI 316 stainless steel or of materials not adversely affected by the electrolyte solutions. The capacity of the reservoir shall be not less than that specified in table I. The reservoir shall be divided into two compartments: one compartment for collecting and settling dirty electrolyte from the machining area; and the other compartment for overflow of settled electrolyte.

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3.4.8.2 Electrolyte pump. The electrolyte pump shall have a capacity conforming to table I, and shall be of the type not affected by abrasive particles in the electrolyte. The pump shall be designed so that the motor and electrolyte seal can be removed without removing the pump from the line. The pump shall be designed to operate continuously without overheating the motor, the pump or the electrolyte flow control valve in the closed position. All components of the pump shall be fabricated of materials not affected by electrolyte solution.

3.4.8.3 Electrolyte filter. The electrolyte system shall have a washable type filter or removable type filter capable of removing all solids larger than 100 micron particle size. Pressure gages shall be provided to indicate the pressure drop across the filter to enable to operator to determine when element replacement or cleaning is necessary. The filter shall be located in an accessible position to facilitate servicing.

3.4.8.4 Piping and hoses. All piping and hoses shall be of adequate sizes to minimize pressure drop in the system. All piping and hoses shall be fabricated of a material not adversely affected by the electrolyte and shall have a burst pressure rating not less than four times the maximum pressure that can be developed in the system. Hoses shall be of adequate length so that they are not stretched or strained when the parts to which they are connected are in their extreme positions, and shall be routed and attached in a manner that will prevent rubbing and chafing. The piping system shall have all necessary regulating valves, relief valves, and shut-off valves required to regulate the flow of electrolyte to the various circuits of the electrolyte system.

3.4.9 Electrical power supply unit. The grinding machine shall be equipped with a direct current power supply unit with its output terminals connected in such a manner that the workpiece will be anodic and the grinding wheel cathodic. The power supply unit shall have a continuous output current rating of not less than 300, 600, or 1000 amperes as specified (see 6.2). The current output shall be automatically variable, depending on the contact area between the grinding wheel and workpiece, as to tend to maintain a constant current density (amp/sq.in.). The power supply unit shall have a continuously adjustable output voltage range of 0 to 15 volts. The unit shall be provided with a voltmeter and ammeter for indicating DC voltage and grinding current.

3.4.9.1 Power supply cabinet. The power supply cabinet shall have removable panels or doors to provide access to the interior components to facilitate trouble shooting and servicing. Doors or panels shall have safety interlocks that disconnect input power to the power supply unit when doors are opened or panels removed.

3.4.10 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A

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filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a cleanout access plate if the reservoir is not removable.

3.4.11 Electrical system. Unless otherwise specified (see 6.2), the electrical system shall conform to NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3 phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.11.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof, enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.11.2 Solid-state components. Solid-state design shall be used throughout for electronic components.

3.4.11.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input power supply by a low-voltage transformer having secondary voltage of 120 volts or less.

3.4.11.4 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.6 Performance. Each machine shall be capable of performing as specified herein.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II.

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3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.8.1 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.10 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Workmanship. The workmanship of the machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as

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otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and

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all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41^{0} to 104^{0} Fahrenheit and 20 to 95 percent relative humidity.

4.8.2 Operational test. Each machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.7. The alignment tolerance test shall be performed prior to and subsequent to the performance test(s) given in 4.8.4 through 4.8.4.1. Any machine adjustments required to comply with 3.7 during or after the performance test shall be cause for rejection.

4.8.4 Performance tests. The machine shall be tested to demonstrate that it is capable of producing a surface finish not rougher than 15 route mean square. The test shall be conducted on a workpiece of AISI 316 stainless steel. The dimensions and configurations of the grinding wheel and workpiece shall be at the option of the machine manufacturer. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.8.4.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The grinding machines covered by this specification are intended for electrolytic surface grinding operations normally required in general machine shop practice.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. If the electro-chemical grinding machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

f. Digital readout system, if required specify axis of travel required (see 3.2.1.2).

g. Specify any special digital readout system features (see 3.2.1.2).

h. Additional safety and health requirements, if required specify configuration (see 3.2.5).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystems class and the emission and susceptibility required (see 3.3.8).

k. Specify power rating of power supply unit to be furnished (see 3.4.9).

l. Electrical system and input power, if different (see 3.4.11).

m. Motor enclosure, if different (see 3.4.11.1).

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n. Worklight, if required (see 3.4.11.4).

o. Size and capacity, if different (see 3.5).

p. Additional performance requirements, if required (see 3.6.1).

q. Specify and fully describe optional equipment, if required (see 3.9.1).

r. Specify applicable repair parts, if required (see 3.10).

s. Lubrication plate, if required (see 3.10.1).

t. Information to be provided on lubrication plate, if different (see 3.10.1).

u. Nameplate, if different (see 3.11.2).

v. Inspection conditions, if different (see 4.3).

w. First article inspection, if different (see 4.4).

x. Quality conformance inspection, if different (see 4.5).

y. Acceptance test, if different (see 4.6).

z. Additional performance tests, if required (see 4.8.4.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

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6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

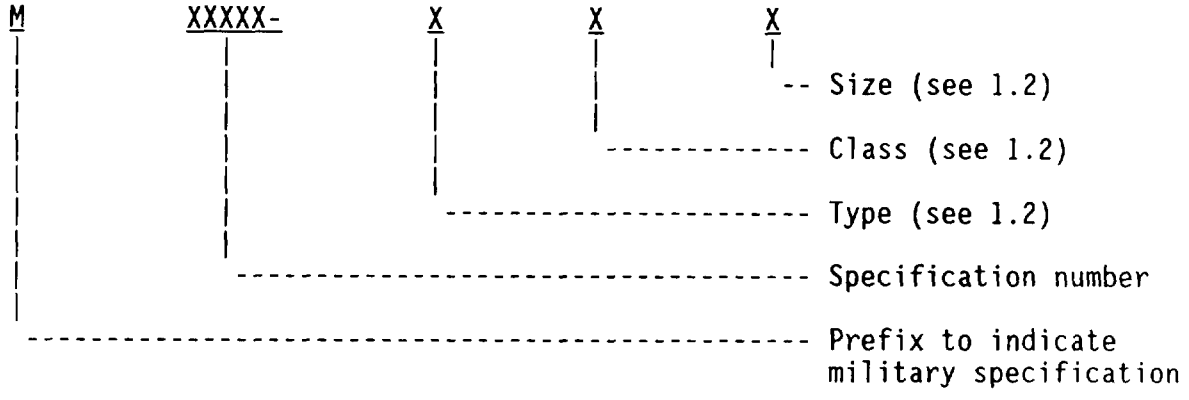
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6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.13 Subject term (keyword) listing.

Base
Column, cast iron
Electrolytic
Grinder
Saddle
Spindle drive
Table, corrosion - resistant

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TABLE I. Sizes and Characteristics

Characteristics	Requirements		
	Size 1	Size 2	Size 3
Worktable area, inches	6x18	8x24	12x36
Worktable travel, inches (min.)	19	25	37
Worktable cross travel, inches (min.)	6-1/2	9	13
Table T-slots, number	1	3	3
Table T-slots, size, inches		.565/ .563	
Bottom of wheel to table			
up, min., inches	15	15	15
down, min.	0	0	0
Wheel size, inches			
diameter, max.	8	10	12
width, max.	3	3	3
Spindle, H.P. minimum	5	5	7-1/2
Spindle speed, rmp	2700	2100	1750
Table motor, gearhead			
H.P., D.C	1/4	1/4	1/4
Table, feed range		1/4 to 20 IPM	
Table, rapid movement rate		40 IPM	
Electrolyte tank			
Cap. gal	20	50	50
Handwheel vertical downfeed			
per revolution, inches, min.	0.050	.050	.050
Handwheel cross feed			
per revolution, inches, min	.125	.125	.125
Power supply, ampere		as specified	
volts, minimum	0	0	0
volts, maximum	15	15	15
Pump motor, H.P.	1/4	1/4	1/4
Pump capacity GPM	10	10	10
minimum	10	10	10
Mist collector, vol.		450 CFM.	

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TABLE II. Alignment Tolerances

REQUIREMENTS	TOLERANCES (Not greater than)
Rise and fall, respectively, of table in its lengthwise movement	.0002" per 18"
Surface of table parallel with its transverse movement	.002 on total width of table
Vertical adjustment of wheel spindle housing, square with table in cross plan of machine	.0004" per 6"
Wheel spindle parallel with table	.0004" per 6"
Wheel spindle - taper and centering sleeve, respectively, of wheel spindle for true running	.0002"
Collet face run out	.0002"
Wheel spindle square with table	.0004" per 12"
Worktable - table flat or level both directions straight edge test diagonally	.0002" per 18"

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Navy - OS, AS
Air Force - 84

User activities:
Navy - SH, MC

Preparing activity:
DLA - IP

Project (3410-0045)

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-80254A	Electro-Chemical Machine, Grinding, Surface

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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